# SKILLBRIDGE: A WEB-BASED OJT PLACEMENT DECISION SUPPORT SYSTEM WITH NLP-ASSISTED SKILLS ASSESSMENT AND COMPANY-POSITION RECOMMENDATION

**A Capstone Project Presented to**

**Faculty of the Institute of Computing**

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# CHAPTER 1

# INTRODUCTION

## Background of the Project

The fast development of digital technologies has brought new ways of job search and recruiting of appropriate candidates. Nowadays, there is an intense competition within the world labor market, which has resulted in a gap between the competencies possessed by graduates and the ones required by employers. The mentioned problem has raised concerns, as fresh graduates often remain without employment in spite of numerous job offers in various industries.

To solve this problem, researchers began to explore the possibility of using intelligent systems that would help to increase efficiency of the process of job matching. As it was proved by Varelas et al., the process of automatic classification of job vacancies is possible with the help of Natural Language Processing (NLP), helping to find correspondence between job advertisements and suitable applicants [1]. In turn, Putka et al. found out that the process of prediction of job-related skills, knowledge and abilities is possible with the help of machine learning models [2].

In the Philippines, the issue of employability is still one problem that cannot be disregarded. Though there has been a noticeable increase in the number of ICT graduates in the Philippines, there are still a lot of them who are underemployed or misaligned with the job they are holding because of their entry level. The traditional techniques being utilized for looking for work such as

participating in job fairs, applying through walk-ins and referral process are not only tiring but also less efficient to the students and potential employers. Based on the study of Mendez and Bulanadi, 83.6% of graduating CICT students prefer online job searching yet they also have a demand for a matching system that would suit their skills with the employers [3]. Also, according to Cabalbag and Ontiveros, the utilization of manual systems for managing student training program is inefficient and error-prone to feedback [4].

In the local context, there is also the same problem for DNSC with making sure that the students get placed into companies that suit their competencies. The students who are enrolled in ICT courses face problems regarding the identification of companies and jobs that suit their competencies, particularly during the transition from academic study to On-the-Job Training (OJT). As a result, the placement is done based on incomplete information and not on the competencies of the students. The works of Permana and Pradnyana indicate that the use of an artificial intelligence-based recommendation system can accurately match students to internships according to their competencies [5].

Moreover, the proposed SkillBridge system is consistent with the CHED A.C.H.I.E.V.E. 2030 Agenda in Thrust C: Centralized One-Nation Human Capital Development Plan, Thrust H: Harmonized SDG-Based Research & Innovation, and Enabler I: Integrated Real-Time Data Collection and Analytics System [6]. In Thrust C, SkillBridge helps to match the skill levels of the students with the demands of the labor market through assessment of their technical skills and suggesting the suitable companies for OJT. In Thrust H, SkillBridge works as an

applied research and innovation activity that helps to solve issues of skills mismatch and OJT placement using ICT. It also supports Enabler I by organizing assessment results, competency profiles, company-position requirements, and recommendation data to provide instructors and OJT coordinators with timely information for evidence-based placement decisions. These alignments reflect CHED's goal of developing future-ready and employable graduates through industry-responsive education, SDG-based innovation, and data-driven higher education services.

The project is consistent with the DNSC SAILS Research, Development, and Extension Agenda for 2026-2030 that represents the name of Davao del Norte State College, Sustaining Gains in Agri-Aqua to Accelerate Innovation and Inclusion Leading to Secured Futures [7]. SkillBridge directly addresses the Strategic Priority 3, which is Innovation and Inclusion (the I² Agenda) promoting ICT for Development and the use of digital technology in bridging institutional and societal gaps. This system employs web-based technology, structured competence assessment, NLP-assisted processing, and recommendation approaches to improve the process of on-the-job training (OJT) placement. On the other hand, SkillBridge helps to meet the objectives of Strategic Priority 5: Secured Futures (the Security Agenda), especially in the areas of future proofing and well-being, since the system facilitates OJT placement that is in line with students' competences and prepares them for the industry requirements. Both strategic priorities are incorporated into one institutional agenda as SkillBridge is

based on the application of innovative technologies and inclusive access to competence and placement information.

At the national research level, SkillBridge corresponds with the DOST Harmonized National Research and Development Agenda 2022-2028, particularly with regard to the Industry, Energy, and Emerging Technology Research and Development Agenda [8]. The relevant priority domains are ICT Innovations and Artificial Intelligence, which contribute to the creation of intelligent and digital systems aimed at fulfilling the needs of institutions, economies, and society. The SkillBridge contributes to such domains through the use of structured competency data, text normalization, text processing using TF-IDF, cosine similarity, data analytics, and the ICT decision-support system in OJT placement.

The present capstone project, SkillBridge, is a web-based OJT placement decision-support tool intended for use at DNSC. The system uses NLP-assisted methods to process students' competencies obtained through structured assessments and compare them with company-position requirements. It then generates a ranked list of placement recommendations to assist students, instructors, and OJT coordinators in identifying suitable companies and positions. The system also considers category-based assessment scores, textual competency information, position requirements, and location as factors in producing recommendations. However, SkillBridge serves only as a decision-support tool, and the final placement decision remains with the authorized instructor or OJT coordinator.

SkillBridge is aligned with the United Nations Sustainable Development Goals of SDG 4: Quality Education, SDG 8: Decent Work and Economic Growth, and SDG 9: Industry, Innovation and Infrastructure [9]. Under SDG 4, the project promotes the growth of relevant technical and vocational competencies by providing a framework in the form of skills assessment and generation of competency profiles. These profiles allow both students and teachers to gauge the strengths and weak zones of the students. Furthermore, by matching assessment results with OJT position requirements, SkillBridge helps ensure that OJT placements are closely aligned with the competencies developed via students' academic training [9].

With regards to SDG 8, SkillBridge facilitates the transition of students from academic training to OJT via competency-based recommendation of company positions for OJT. SkillBridge helps students find OJT positions that match their competencies, hence giving students a more relevant workplace experience as well as making it easier to apply their academic knowledge in practice and prepare for their future work life. While the project does not provide guaranteed placements and does not assign any companies to students, the aim of the data-driven recommendations provided is to help instructors and OJT coordinators make more informed and objective decisions in placing students [9].

As far as SDG 9 goes, the project makes a contribution through the development of an innovative ICT solution based on structured assessment, NLP-assisted processing, similarity computation, and data analytics applied to an existing institution's process [9]. Rather than using manually collected records,

generalized student data, and available company slots, SkillBridge transforms students' competencies and company-position requirements into structured data usable in decision-making. Also, SkillBridge enhances the cooperation of industry and academia by allowing direct comparison of students' competencies with those of partner companies and OJT positions. This is in keeping with the CHED Harmonized SDG-Based Research and Innovation Thrust and DNSC I² Agenda which promote research and innovations in response to practical needs [6], [7].

Despite the availability of online job-matching platforms, internship recommendation systems, and OJT record-management applications, an important gap remains in the integration of competency assessment, student skill profiling, company-position requirement analysis, and placement recommendation within a single institution-based system. Existing systems commonly focus only on job searching, trainee monitoring, record management, or recommendation and do not provide a unified platform that assesses DNSC students' competencies, converts assessment results into structured competency profiles, compares those profiles with company-position requirements through NLP-assisted and similarity-based techniques, considers location as a practical placement factor, and produces ranked recommendations for OJT decision-making. Therefore, there is a need to develop SkillBridge as a web-based, competency-driven, and data-supported OJT placement decision support system specifically designed for the DNSC Institute of Computing.

## Objectives of the Study

This research proposes development of a web-based OJT placement decision support tool called SkillBridge which uses NLP techniques to extract information from student competency profiles and produce ranking of suitable companies and positions. The goals of the study are to:

1. Develop a web-based OJT placement decision support tool which would allow storing student profiles, company information, position description, competency tests scores and OJT placements recommendations.
2. Create and implement a skills test module which would help instructors create, edit and deploy a questionnaire, analyze students' answers and classify them by predefined skill area.
3. Select the most appropriate and efficient pretrained Natural Language Processing models among three (spaCy en_core_web_sm, spaCy en_core_web_md and Stanford Stanza English) which will be used for pre-processing competency profiles of students and companies/positions.
4. Implement a recommendation engine using techniques such as similarity between category-score values, text similarities obtained through NLP techniques and location similarities to find matching companies/positions for particular student.
5. Implement a data analytics and reporting tool for visualization of statistics about student competency profiles, recommendation results, matches between student and company/position, free OJT placements and other relevant information.

## Significance of Study

The SkillBridge project employs a data-based technique of OJT placement that has changed the manual process into an organized one. The process ensures effective decision making, eliminates evaluator's bias, and thus becomes more effective in placing the students in practical situations. The below list shows the key stakeholders who benefit from this system, listed in descending order of importance:

**Students.** The main beneficiaries of SkillBridge are the BSIT students because they will be the ones to use it. Using SkillBridge, there will be assessments done digitally that will create an accurate competency profile based on the student's skill level in the respective IT areas. There is also a recommendation part where the matching of students to suitable OJT companies and positions is done.

**OJT Coordinators & OJT Instructors**. The instructors will gain by having lesser administrative burden due to the automated testing and assessment of competencies and placement of trainees. Consistent and standardized information about the competencies of trainees will enable proper placement of the trainees.

**Educational Institutions (DNSC Institute of Computing).** The educational institutions will gain by having a technologically-based management of OJT that will promote better relations with the industries, maintain

standardized placement records, and provide competency information that may help improve the curriculum.

**Industrial Partners**. Companies would greatly benefit from the intake of students who possess the right skill sets necessary for the job, as this would minimize the training period and enable companies to give tasks that are suitable for the students' skill sets.

**Future Researchers**. This paper has provided an example of how NLP could be applied to educational decision support systems. This research could serve as a guide to future researchers in related fields.

**Scope and Limitation**

The SkillBridge is a web-based OJT placement decision-making tool created for implementation at the Institute of Computing of the Davao del Norte State College located in Panabo City, Davao del Norte, Philippines. The tool caters to three types of users: administrators, instructors, and students using institutional Google accounts (@dnsc.edu.ph). This software is built around three main components: (1) a skills assessment component allowing creation of questionnaires by the instructors within certain categories of competencies, (2) an NLP-based competencies profiling component analyzing answers to the questionnaire in order to develop a profile of the learner with scores and skills tagged, and (3) a recommendation component generating the recommendations based on the company job requirements.

The system is utilized merely as a decision-support tool and does not carry out any automated decisions regarding placement, which are made either by the instructors or OJT coordinators. The NLP module works strictly with structured assessment data and does not make use of any external datasets or more complex generative AI tools. It does not interface with any external job websites, government employment databases or HR management systems of firms and works only internally for one particular organization. The precision of the recommendations is influenced by the quality of assessment design and data provided by the firm, and the evaluation consists in usability, recommendation precision and user satisfaction.

## Review of Related Literature and Works

The following section presents an overview of the existing literature on the topic of job matching, competency modeling, and recommendation system. This is done in order to critically examine the existing studies on this issue, outline the key issues, and point out the research gap that would justify the development of the proposed system. It becomes increasingly important to integrate NLP, recommender systems, and DSS.

### Related Literature

NLP has contributed to the development of automated processes for matching jobs and extracting skills. Classification of job vacancies into structured occupational categories using NLP makes matching efficient [1]. Moreover, NLP models have the ability to accurately extract knowledge, skills, abilities and other characteristics (KSAOs) relevant to jobs [2].

The latest progress in this field highlights the usefulness of applying deep learning along with NLP in recruitment systems. Recent empirical studies show that such systems achieve higher accuracy in job recommendations using textual data from resumes and job descriptions, thus allowing to transform unstructured text data to structured skills [10], [11].

Recommender systems are used widely on employment websites to suggest jobs tailored to the interests of the user. Recommenders use collaborative filtering to suggest the positions which are preferred by the users[3].

Still, more accurate recommender systems and the solutions for the problem of cold-start can be achieved using hybrid methods, including content-based filtering and machine learning [12], [13].

DSS helps to make better decisions through the use of various data sources and models of analysis. They help to make placement decisions by providing objective information grounded on data analysis [14]. Many implementations of DSS are not equipped with effective data processing mechanisms.

There are also applications of artificial intelligence in internship placement systems. AI methods can be used for an accurate placement of students in internships according to their competencies [5]. Yet, many systems do not provide any opportunities for their integration and scaling.

## Synthesis of Key Themes

From the analyzed literature, the following main ideas can be distinguished. Firstly, NLP techniques are widely used for competencies extraction and classification that allows automatic translation of resumes and job descriptions into structured competencies data [1], [2]. However, despite its effectiveness, many approaches are still limited to text analysis and are not incorporated in larger systems.

Secondly, recommender systems are important for job matching as they provide personalized recommendations. Collaborative filtering works through users' interactions and behavior, whereas hybrid approach improves efficiency and solves some problems, such as cold start problem [3], [12], [13]. However, most of the systems do not consider competencies-based job matching as their key objective.

Thirdly, decision support systems improve the job placement process as they analyze data from different sources and generate data-driven recommendations [14]. However, despite these advantages, many systems still lack advanced analytical and intelligent processing abilities.

### Identification of Gaps

Nevertheless, despite all achievements of the above-mentioned technologies, there are some gaps in the current literature as well as corresponding systems. First of all, many works do not use NLP-based competency profiling together with recommendation systems resulting in the low-quality matching process. Also, most of the systems are based mainly on users’ interactions without considering their skills and competences. Finally, there is no number of fully integrated systems which would include Natural Language Processing, recommendation technique, and decision support system.

These gaps indicate the need to create a platform that will incorporate the intelligent data processing, competency-based matching and decision support in order to improve the placement process. Based on the reviewed literature,

Natural Language Processing, recommendation technique, and decision support systems proved to be useful in the job and internship matching process. However, the current systems are still limited by the lack of proper integration and the competence-based approach. Therefore, these results confirm the necessity of development of the above-mentioned system called SkillBridge.

**Related Works**

Current systems use different approaches to match jobs and manage internships. The Job Matcher system employs collaborative filtering to provide job suggestions according to user preferences, but there is no competency assessment [3]. At the same time, the Management Information System for Tracking On-the-Job Trainees focuses on trainee record tracking, but there are no intelligent suggestions in it [4].

The AI-based internship recommendation system uses competency assessment using artificial intelligence, thus having enhanced accuracy, but no system integration [5]. Some other systems, which use deep learning and natural language processing, improve accuracy of resume-to-job matching, but they often lack assessment and decision-making capabilities [10], [11].

All the above-mentioned systems demonstrate the advantages of artificial intelligence and recommenders, but still fail to provide an integrated system with all the necessary components.

**Table 1.** Comparative Analysis of SkillBridge and Related Works

| System | Job / Internship Recommendation | OJT Monitoring | Competency-Based Analysis | NLP-Based Profiling | Intelligent Recommendation | System Integration | Decision Support System (DSS) |
|---|---|---|---|---|---|---|---|
| Job Matcher [3] | ✓ | | | | ✓ | | |
| OJT Tracking System [4] | | ✓ | | | | | |
| AI Internship Recommender [5] | ✓ | | ✓ | | ✓ | | |
| NLP Job Recommendation Systems [7], [8] | ✓ | | ✓ | ✓ | ✓ | | |
| Proposed SkillBridge System | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

## Definition of Terms

The following terms are defined for use in the context of this study:

**Administrator.** An account user of SkillBridge charged with the responsibility of managing accounts, company information, job positions, and system configurations.

**Assessment Questionnaire**. A set of assessment questions designed by instructors in SkillBridge, which is grouped into competency categories and is used in a digital form to assess students' skills associated with OJT.

**Competency Category.** A specific domain of technical skill associated with BSIT program, like Web Development and Networking, which is used for classifying assessment items and organizing students' competencies.

**Competency Profile.** This is an automatic profile of competencies produced as a result of the outcomes from assessments, and includes numbers and tags of competencies based on organizational needs.

**Decision Support System (DSS).** Information system to assist the process of decision making for instructors and coordinators in placing trainees in OJT programs.

**Instructor.** A SkillBridge system user designing assessment questionnaires, monitoring the performance of the students, and using system-generated results to make OJT placement decisions.

**Natural Language Processing (NLP)**. The approach used in SkillBridge to analyze the assessment results and to extract skill information and structure the competency profiles.

**On-the-Job Training (OJT).** The academic program, in which students are assigned to partner companies to obtain practical experience and apply theoretical knowledge acquired through learning.

**Placement Recommendation.** The ranked list of company and position matches generated by the system on the basis of the alignment between student's competency profile and company needs.

**Recommendation Engine.** The system module generating the priority-based OJT placement recommendations based on students' competency profiles and company needs.

**Role-Based Access Control (RBAC).** System feature that offers restricted access to features and data depending on the user role (administrator, instructor, and student).

**Student.** A SkillBridge system user who completes the assessments, views competency profiles, and receives OJT placement recommendations.

# CHAPTER 2

## METHODOLOGY

The project will implement the hybrid Project Management Approach that combines the waterfall structure with the Agile approach to development (see Fig. 1). Specifically, the waterfall structure will be used in the first phases of the project – the planning phase, the requirements analysis phase, and the system design documentation phase – as the outputs should be defined and verified prior to the actual development. Further, the Agile development method will be used in the implementation and testing phases.

The rationale behind the choice of the hybrid approach is linked to the fact that the system under development has both static and dynamic elements. The requirements concerning system planning and design are very important and require proper planning and documentation, while the aspects of intelligent recommendations and UI are easier to develop and test.

The project will be developed based on the PADIM framework, which includes five phases: planning, analysis, design, implementation, and maintenance. Each of these phases is an important part of the SkillBridge development.

At the Planning stage, proponents will define the project scope, goals, schedule, and resource requirements according to the accepted concept proposal. The proponents have determined the current problem in the DNSC

organization: the lack of any structured and automated method to match the IT students with appropriate OJT placements. At the planning stage, the feasibility of development of a web-based Decision Support System with a feature of skill matching using NLP technologies will be assessed considering the schedule and resources available for the project.

At the Analysis stage, the proponents will identify and document all system requirements via reviews of the current DNSC OJT process, interviews of stakeholders, and identification of roles and needs of the users. In addition, proponents will determine functional and non-functional requirements of the system.

At the Design stage, requirements obtained at the Analysis stage will be used to develop system architecture, which will include database structure, prototypes of user interfaces, data flow diagrams, entity-relationship diagram, and selected NLP algorithms and recommendations techniques.

At the Implementation stage, proponents will follow the iterative approach, in which coding, integration, and testing will be performed in weekly development cycles. Such approach allows the continuous evaluation and improvement of system components in view of the test results and feedback from users.

Finally, at the Maintenance stage, emphasis will be placed on system evaluation and testing, User Acceptance Testing, issues solution, optimization, and maintenance planning.

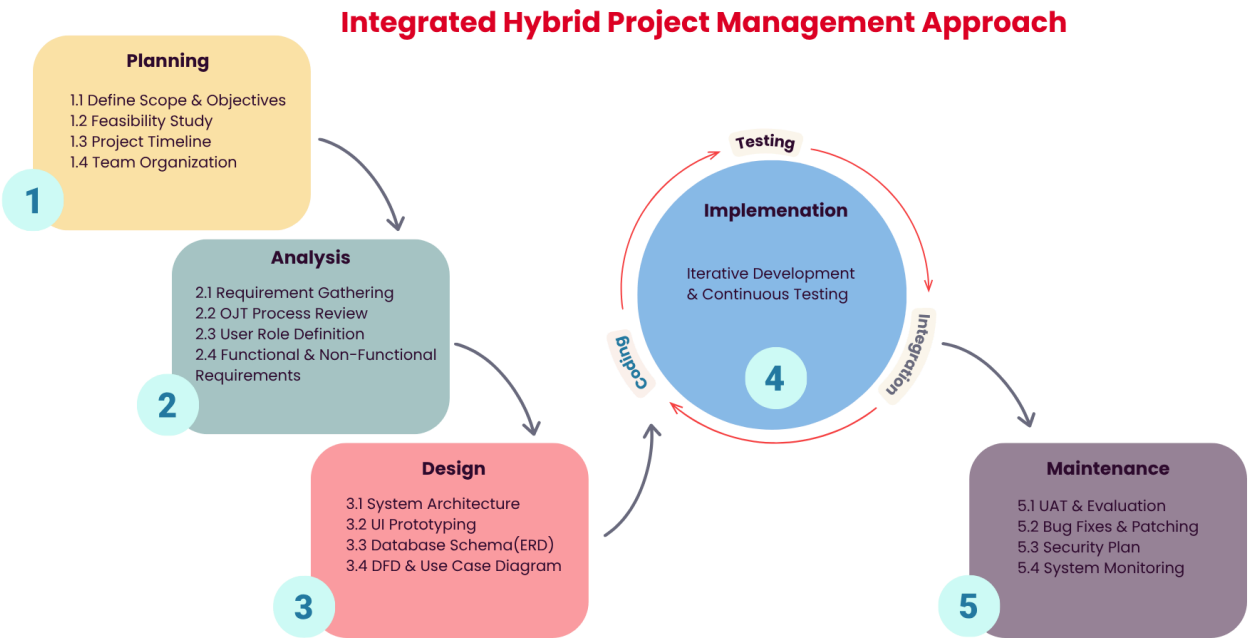



**Figure 1.** Integrated Hybrid Project Management Approach (PADIM Framework)

## SYSTEM PLANNING

### Project Team Organization

The project team shown in Figure 2 is arranged in such a way as to guarantee proper coordination and clear delineation of tasks within it. The Adviser acts as an overseer and gives guidance on a general level to both academic and technical issues involved in the project and ensures its compliance with the institution's standards. The Project Manager directs planning, coordination of tasks, development work, and progress tracking so as to make sure that the project is delivered in time. Other participants include the System Analyst and Documentation Specialist who undertake the role of requirements analysis and prepare the necessary documentation and the System Developer and Documentation Specialist who help in development and are also the co-authors of the capstone paper.

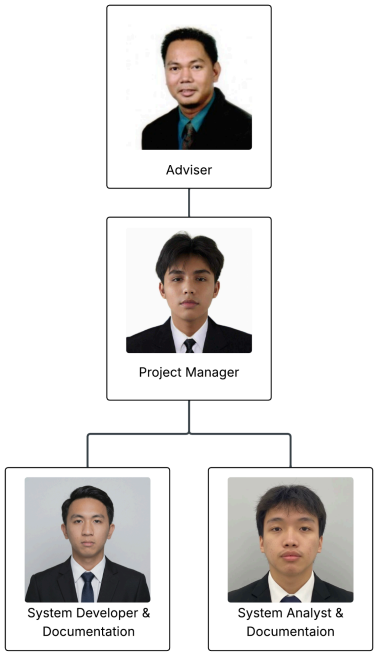



**Figure 2.** Project Management Team Organization

**Work Breakdown Structure**

The Work Breakdown Structure (WBS), as shown in Figure 3, is an illustration of the phases in which the project is broken down based on the PADIM model. The WBS clearly specifies the activities to be carried out in every phase of the development process of the system, right from Planning and Analysis up to Design, Implementation, and Maintenance phases. The process provides an opportunity for the proponents to effectively manage and monitor the implementation of the project. Project phases are operationalized in the form of different tasks, which include requirements analysis, architecture design, implementation, testing, and evaluation.

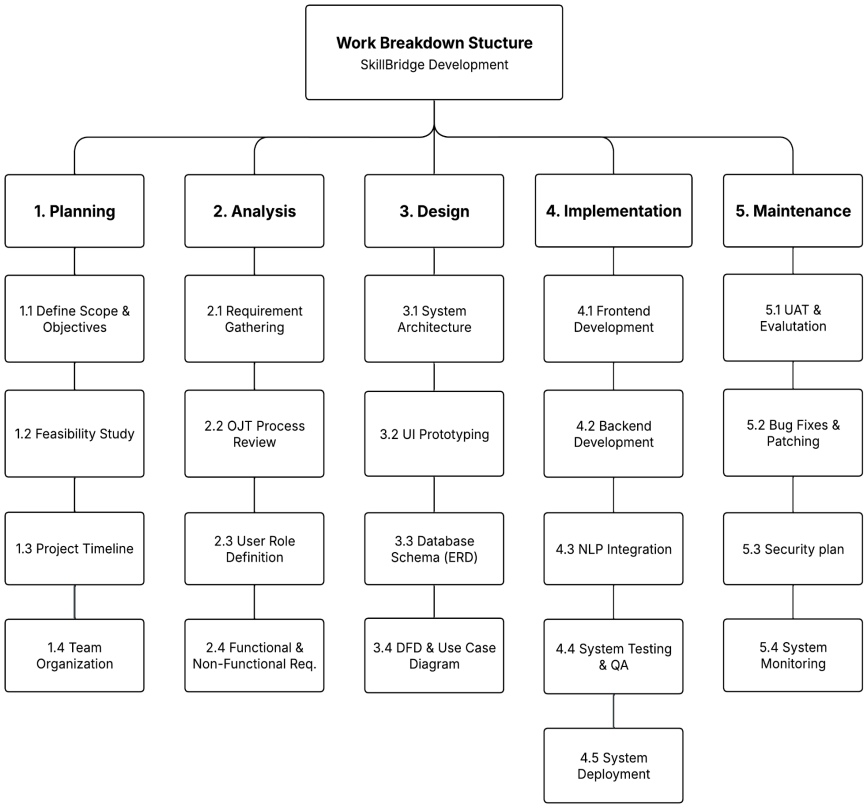



**Figure 3**. Work Breakdown Structure.

## Gantt Chart

The Gantt chart illustrated in figure 4 indicates the proposed schedule of activities that would be involved in the creation of SkillBridge within a period of seven weeks from April to May 2026. The chart illustrates the process and the expected time frame for the completion of each stage of development that would match the project requirements. The stages include front-end development, back-end setup, NLP incorporation, wiring, and final deployment of the platform. The chart acts as a reference tool for advocates who wish to track their progress.

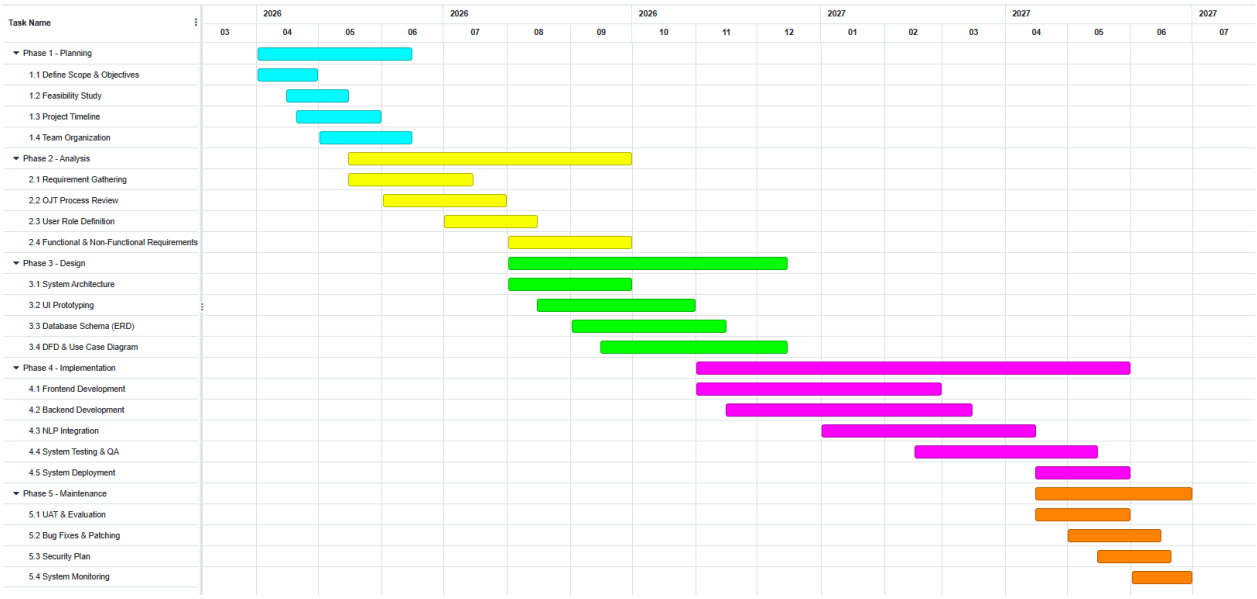



**Figure 4.** Gantt Chart of the system.

## SYSTEM ANALYSIS

### System Architecture

SkillBridge system architecture illustrated in Figure 5 follows the three-tiered client-server model that provides an understanding of the interaction between various system components in order to perform the skills assessment, matching, and placement recommendation for OJT. This model consists of three major layers including the Presentation Layer (frontend), Application Layer (backend), and Data Layer (database).

At the Presentation Layer, the React 19 framework that uses Vite and styling in Tailwind CSS 4 is used to create a system. There are 16 pages designed for three different users and hosted by Vercel for global distribution via content delivery network. The communication with the backend is carried out using REST API with the help of Axios using JWT Bearer tokens. The frontend can be accessed at https://skill-bridge-six-psi.vercel.app.

At the Application Layer, the Python framework Django 6.0.4 with Django REST Framework 3.17.1 is used to implement the system. This layer handles all of the business logic such as user authentication, tests' management, auto-grading with the help of natural language processing, and recommendations using cosine similarity. The backend is hosted on Railway.app and can be access at https://skillbridge-production-1e3c.up.railway.app.

At the Data Layer, the PostgreSQL is used via Supabase which is a managed service providing a PostgreSQL database with automated backups and

connection pooling. There are 14 tables in total which belong to four major categories: User Management, Assessment Engine, Response and Scoring, and Placement Matching. All the data of the system, including profiles of users, questions of the test, students' answers, skills scores, companies, and recommendations, will be stored and managed here.

In addition, the system utilizes a number of external services. For student login, Google OAuth is used but only @dnsc.edu.ph accounts are allowed. The PSGC API is used to create cascading address selections for profiles and corporations. The company address are geocoded with the use of Nominatim, and Excel sheets that can be parsed by SheetJS.

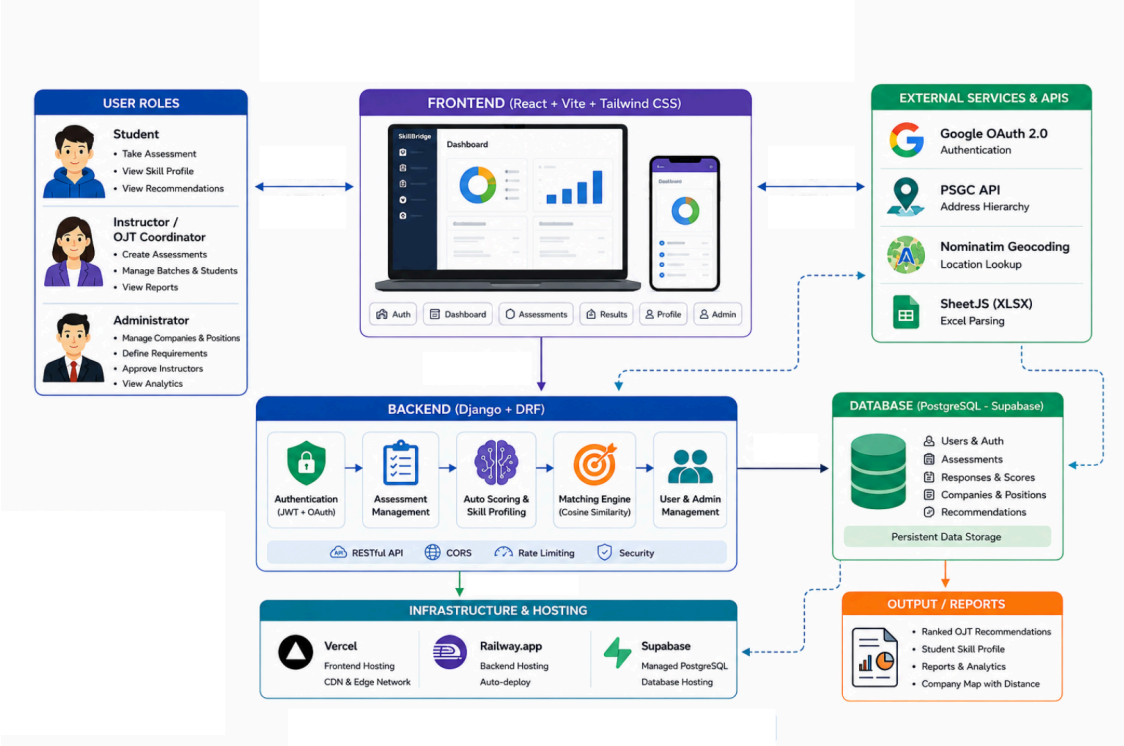



**Figure 5.** System architecture of the project.

## Functional and Non-Functional Requirements

The functional and non-functional requirements outline the requirements in terms of the expected functions and qualities of SkillBridge. The functional requirements entail the main functionalities that will be provided by the system, namely authentication, assessment management, automated scoring, recommendations creation, and dashboard visualization for all the user roles. On the other hand, the non-functional requirements describe the quality requirements that the system needs to fulfill according to the ISO/IEC 25010 standard, and they include such areas as reliability, usability, security, and performance efficiency.

### Functional Requirements

The system must have the following functionalities:

1. The system must provide secure authentication and role-based access control for students, instructors, and administrators. It must provide Google OAuth based authentication for student accounts and email-password based authentication for staff accounts.
2. The system must enable students to manage their profile information, which includes course information, school identification number, contact information, address, and location information.
3. The system must enable instructors to create and manage batches of students, to enroll students using email ids and to automatically create student accounts for unregistered student email ids.

4. The system must enable administrators to manage company profiles, company locations, OJT positions, OJT slot details, required percentage of skill categories and position-specific tags.
5. The system must enable instructors to create, manage and deploy competency based skill assessment questionnaires using Multiple Choice, True/False and Identification question types and enable bulk Excel based question import.
6. The system must automatically provide skill category tags for assessment questions using TF-IDF text analysis.
7. The system must enable students to answer competency assessments assigned to them and automatically grade the responses provided by students using correct answer flags for objective questions and NLP text normalization for Identification questions.
8. The system must compute the percentage skill scores in different skill categories for each student and provide structured skill profile information.
9. The system must automatically generate competency profile information for students and rich text description of companies and positions using assessment scores, required percentage of skill categories, skill tags and position specific tags.
10. The system must pre-process the competency profiles and company-position descriptions using the chosen NLP model, perform TF-IDF feature extraction and compute NLP based text similarity.

11. The system must compute OJT placement recommendations considering category-score based similarity, NLP-assisted text similarity and location based similarity.
12. The system must display ranked OJT placement recommendations to students including recommended company, position, match score percentage, distance information and company location on an interactive map.
13. The system must provide role-specific dashboards and reports to students, instructors, and administrators including competency summary, batch statistics, recommendation outcomes, available OJT slots, company position matching information and placement related insights.

**Non-Functional Requirements**

Following are the system's non functional requirements:

1. **Performance Efficiency.** The system will perform and display assessment results and recommendation with response time having less latency. Both auto-scoring and recommendation will be done within a single HTTP request cycle.
2. **Reliability.** To ensure reliable continuous operation with minimal downtime, the system will be deployed using cloud platforms such as Railway, Vercel, and Supabase where automatic failover will also be used.

3. **Usability.** The system will have a web-based user interface which is easily usable by students, instructors, and admins without any need for specialized knowledge in using technical stuff.
4. **Security.** The system will secure its data through authentication using JWT, role-based access control, HTTPS encryption, API rate limiting and restricted Google OAuth domain for student login.
5. **Compatibility.** The system will be available on various devices including desktops, tablets and mobile phones through web interface supporting modern browser without any plugin.
6. **Maintainability.** The system will be written in such a way that it becomes modular and easy for updating, debugging and adding features.
7. **Scalability (Portability/Adaptability).** The system will allow adding of new categories, companies, positions, students and assessments without compromising the performance and recommendation accuracy.
8. **Availability.** The system will be available to the authorized user at any point in time through the use of cloud infrastructure. The user session will remain active through 7 day long JWT refresh tokens.

**Use Case Diagram**

The main actors in the SkillBridge use case, as depicted in Figure 6, include the Administrator, Instructor (OJT Coordinator), and Student. The responsibilities of the Administrator include managing the system, which involves managing users' accounts, student records, company records, job positions, and system configuration.

The Instructor performs tasks related to the creation of questionnaires for skill assessment, managing assessment questions, conducting assessments, accessing student profiles, examining competence profiles, and analyzing placement recommendations.

The Student actor uses the system through logging into the system for management of his/her account and undertaking the skill assessments. In addition, he/she accesses the competence profile and placement recommendations.

The system analyzes the students' results on the skills assessments and makes recommendations to the student on company positions, ranked according to the computed competence score. Despite SkillBridge having the capability of making intelligent recommendations that can help the OJT Coordinator make placement decisions, such decisions still lie in the hands of the OJT Coordinator.

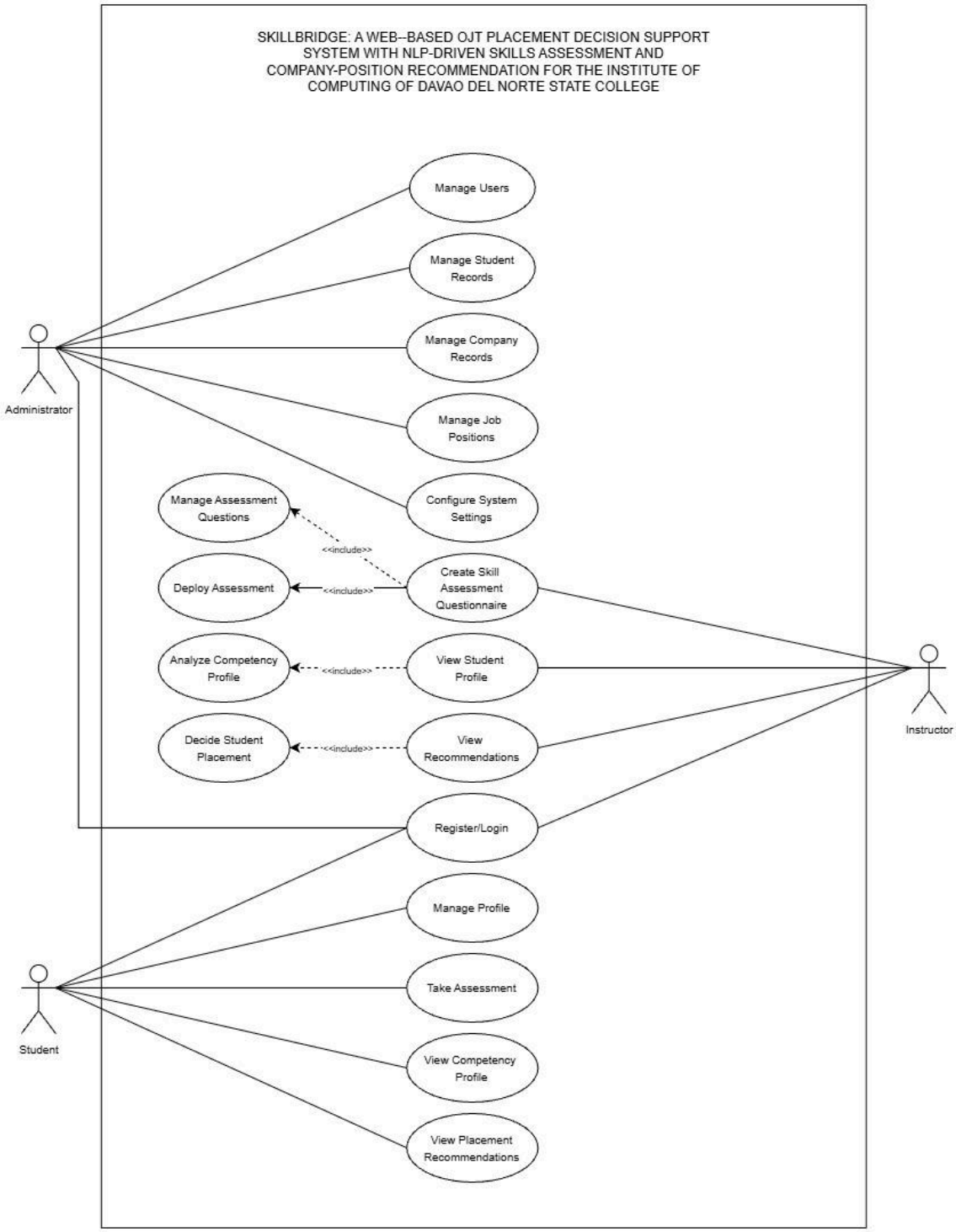



**Figure 6.** Use Case Diagram of SkillBridge

**Context Flow Diagram**

The Context Flow Diagram, depicted in Figure 7 below, depicts an overview of the SkillBridge system, showing how the data flow between the system and the main outside actors: the Student, Instructor (OJT Coordinator), and Administrator.

The Students use the system by providing login and assessment data and receive scores and recommendations for placement as generated by the system according to their competency level and assessment results.

The Instructors use the system by feeding it assessment and batch data and get the generated performance reports and recommendations that help them assess the competency level and suitability for OJT of the Students.

The Administrators provide the system with company and position data and decisions and also manage the system. As a result, they receive analytics reports from the system along with some recommendations.

Thus, the context diagram demonstrates how the SkillBridge system serves as a platform for making decisions regarding assessment data and OJT-related data.

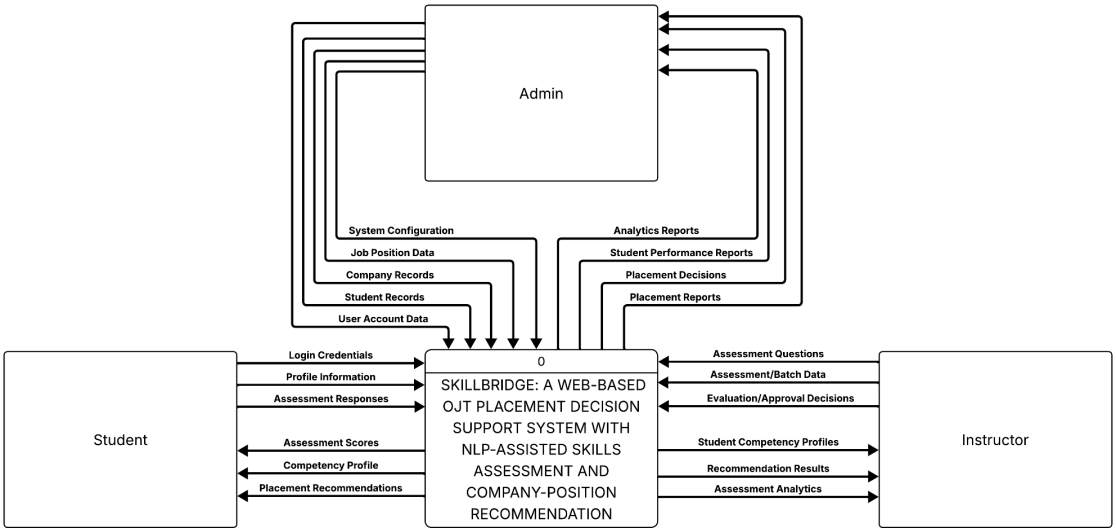



**Figure 7.** Context Flow Diagram (Level 0 DFD) of SkillBridge

**Data Flow Diagram**

The Level 1 DFD (see Figure 8 below), based on the Context Diagram above, represents a decomposition of the SkillBridge system into its main processes keeping the same external entities and data flows. The diagram depicts how data flows through the system as the users interact with it.

Firstly, the process called Authenticating Users manages user's login data and authentication of the access and stores/retrieves user data in the Users data store. Secondly, the Managing Assessments process helps instructors to create and manage assessment data that is then stored in the Assessments data store.

Thirdly, the process named Managing Enrollments manages enrollment data of students, storing enrollment data that will be used further in the system.

Fourthly, the Executing Assessments process retrieves questions for assessments, stores answers to the questions provided by the students in the Responses data store.

Fifthly, the Managing Companies and Positions process manages company profiles and OJT position requirements that will be used in the next process. The process named Processing Recommendations processes the assessment data, answers provided by the students and company-position data and produces competency-based placement recommendations that are stored in the Recommendations data store.

Lastly, the Report Generation process uses data from enrollments, responses and recommendations data stores to generate reports, analytics and recommendation summaries. Thus, the Level 1 DFD describes how SkillBridge processes assessment information and provides OJT placement recommendations based on the data.

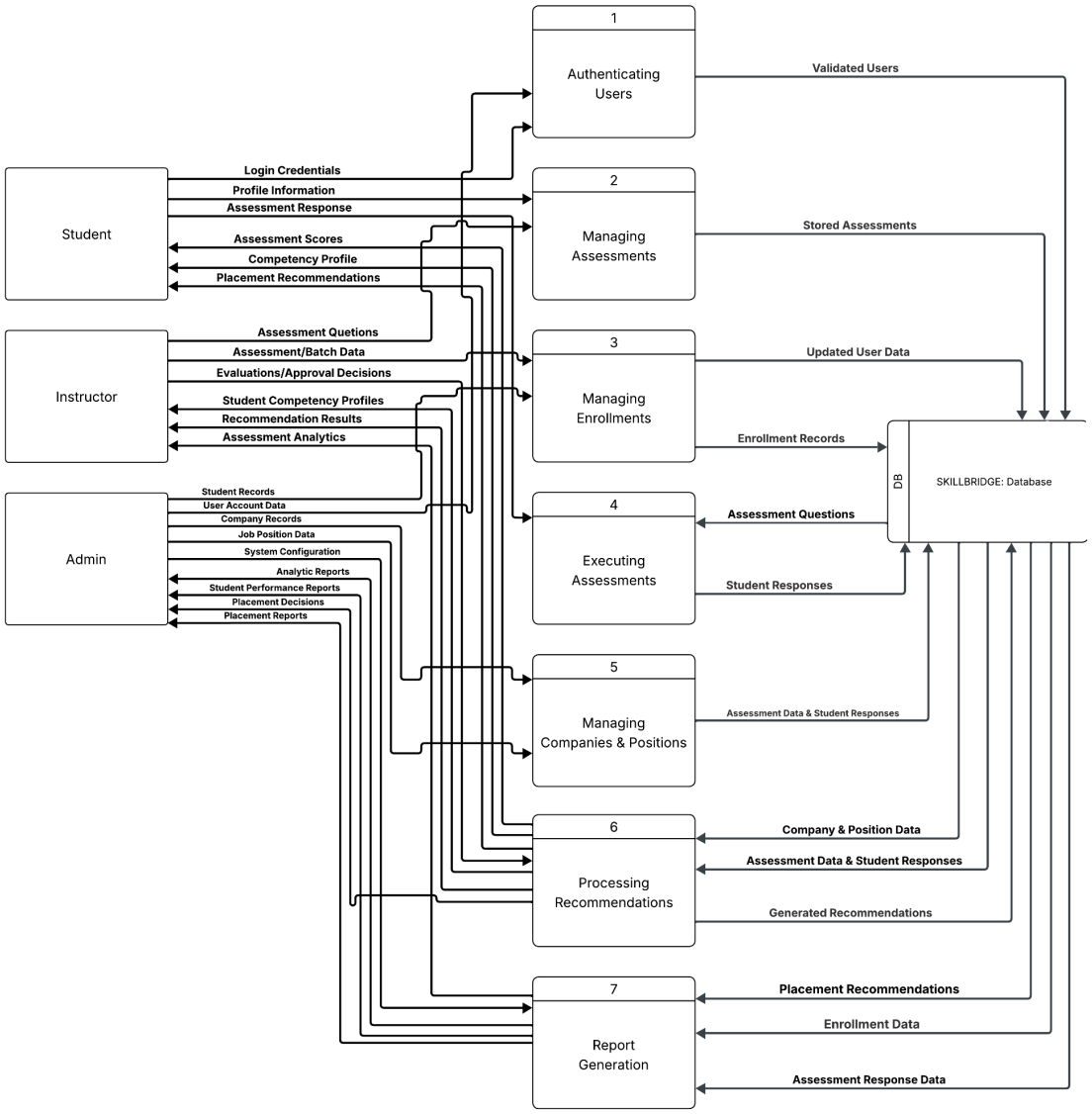



**Figure 8.** Level 1 Data Flow Diagram of SkillBridge

## SYSTEM DESIGN

### Entity Relationship Diagram

In Figure 9. The ER diagram depicts the architecture of the database of SkillBridge. It is illustrated how the data entities are organized within the system and interconnected with each other.

The database contains 14 tables and they are divided into 4 main functional areas: User Management, Assessment Engine, Response and Scoring, and Placement Matching.

User Management area contains tables that manage user accounts for students, instructors, and administrators, as well as manage student grouping and batch enrollment records: api_user, api_batch, and api_batchenrollment.

Assessment Engine area contains tables that allow creating and managing skill based assessment questionnaires: api_assessment, api_question, api_answerchoice, and api_skillcategory.

The Response and Scoring area contains tables that contain student assessment sessions, submitted by them answers, competency scores for each skill category: api_studentresponse, api_responseanswer, and api_skillscore. The table api_responseanswer is a transaction table that creates the relation between the submitted answers and the chosen answer choice.

The Placement Matching area contains tables that manage company profiles, internship positions, required skill percentages per position, and recommendation results that are generated based on match scores: api_company, api_position, api_positionrequirement, and api_recommendation.

Relationships between the entities are created through the use of primary and foreign keys to keep data consistency. For example, one assessment could have many questions and one question could have many answer choices. Also, one student could submit many responses and one student could receive many recommendation records, depending on the computed match score with the positions of different companies.

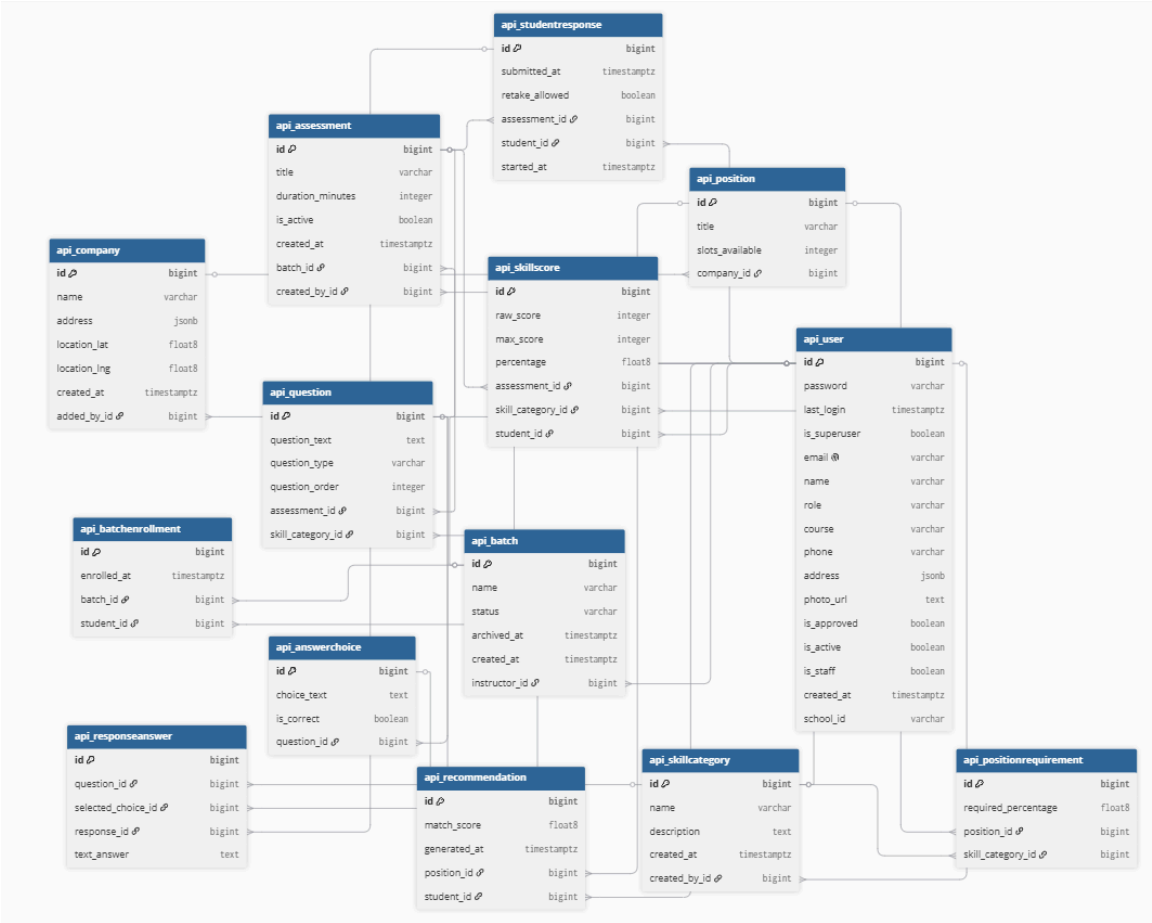



**Figure 9.** Entity relationship diagram of the system

**JSON Schema Diagram**

In order to capture flexible and structured data, JSON fields will be incorporated in SkillBridge. There will be two primary entities that will use JSON fields: the User and the Company tables. The User.address will hold the address of the student's home, address of where the student is staying, and the map pin latitude and longitude in the form of JSON with fields such as stayingAt, travelWilling, province, city, barangay, pinLat, and pinLng. It is the basis of distance calculation feature on the student dashboard. For the Company.address, it will contain the province, city, and barangay of the company, and it will also be used together with the latitude and longitude for map rendering through Nominatim geocoding.

The usage of JSON fields allows for flexibility in the design of the database. Since there are different structures of the address depending on user type and location, it will allow the system to adapt. There is a validation for certain fields like the province and the city, which are required.

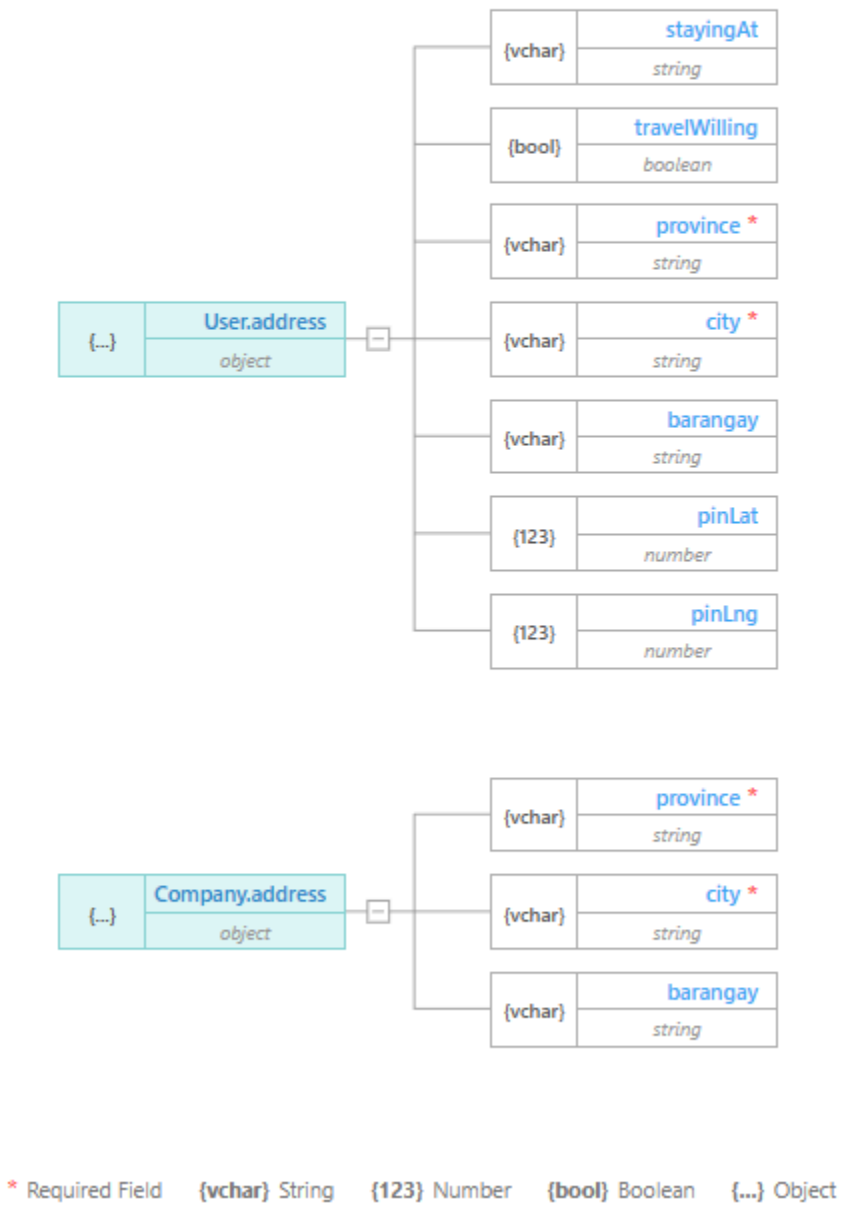



**Figure 10.** JSON Schema Diagram of SkillBridge.

**Data Dictionary**

The Data Dictionary is the document that contains detailed information regarding the main data elements that will be used in SkillBridge. The tables, fields, data types, constraints, and descriptions are defined here to facilitate consistency and comprehension of the way the data is going to be handled in the system. The current section serves as a guide during the implementation, maintenance, and future development of the system. Table 2 contains the list of the most important fields from the most critical tables.

**Table 2.1** User

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique user identifier; auto-generated serial. |
| email | VARCHAR | 254 | Unique Not Null | User's email address; used as login credential |
| password | VARCHAR | 128 | Not Null | Hashed password stored using Django's PBKDF2 algorithm. |
| name | VARCHAR | 150 | Not Null | Full name of the user. |
| role | VARCHAR | 20 | Not Null | Role designation: student , instructor , or admin |
| school_id | VARCHAR | 30 | Nullable | Institutional ID number assigned to students or faculty. |
| course | VARCHAR | 100 | Nullable | Degree program or course of the student (e.g., BSIT, BSCS). |
| phone | VARCHAR | 20 | Nullable | Contact phone number of the user. |

| | | | | |
|---|---|---|---|---|
| address | JSONB | — | Nullable | Structured address in JSON format containing home/boarding location (province, city, barangay), travel preference, and GPS pin coordinates. |
| photo_url | TEXT | — | Nullable | URL pointing to the user's profile photo stored in cloud storage. |
| is_approved | BOOLEAN | 1 | Not Null | Indicates whether an admin has approved the account |
| is_active | BOOLEAN | 1 | Not Null | Indicates if the account is active |
| is_staff | BOOLEAN | 1 | Not Null | Grants access to the Django admin interface when TRUE. |
| is_superuser | BOOLEAN | 1 | Not Null | Grants full unrestricted system permissions when TRUE. |
| last_login | TIMESTAMPTZ | — | Nullable | Timestamp of the user's most recent login session. |
| created_at | TIMESTAMPTZ | — | Not Null | Timestamp of when the user account was created; auto-set on insert. |

**Table 2.2** Batch

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique batch identifier; auto-generated serial. |
| name | VARCHAR | 150 | Not Null | Descriptive name of the batch (e.g., "BSIT 4A – 2nd Semester 2025"). |

| | | | | |
|---|---|---|---|---|
| status | VARCHAR | 10 | Not Null | Current batch lifecycle status: active or archived . |
| archived_at | TIMESTAMPTZ | — | Nullable | Timestamp recording when the batch was archived; NULL if still active. |
| created_at | TIMESTAMPTZ | — | Not Null | Timestamp of when the batch was created. |
| instructor_id | BIGINT | 8 | Foreign Key Not Null | References User.id |

**Table 2.3** Batch Enrollment

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique enrollment record identifier; auto-generated serial. |
| enrolled_at | TIMESTAMPTZ | — | Not Null | Timestamp recording when the student was enrolled into the batch. |
| batch_id | BIGINT | 8 | Foreign Key Not Null | References Batch.id |
| student_id | BIGINT | 8 | Foreign Key Not Null | References User.id |

**Table 2.4** SkillCategory

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique skill category identifier; auto-generated serial. |
| name | VARCHAR | 100 | Not Null | Name of the skill category (e.g., "Web Development", "Communication Skills"). |
| description | TEXT | — | Nullable | Extended description of the skill category and its scope. |
| created_at | TIMESTAMPTZ | — | Not Null | Timestamp of when the skill category was created. |
| created_by_id | BIGINT | 8 | Foreign Key Nullable | References User.id |

**Table 2.5** Assessment

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique assessment identifier; auto-generated serial. |
| title | VARCHAR | 200 | Not Null | Descriptive title of the assessment (e.g., "Midterm Skills Exam – Batch A"). |
| duration_minutes | INTEGER | 4 | Not Null | Time limit for the assessment in minutes |
| is_active | BOOLEAN | 1 | Not Null | Indicates whether the assessment is currently open for student submissions. |

| | | | | |
|---|---|---|---|---|
| created_at | TIMESTAMPTZ | — | Not Null | Timestamp of when the assessment was created. |
| batch_id | BIGINT | 8 | Foreign Key Nullable | References Batch.id |
| created_by_id | BIGINT | 8 | Foreign Key Not Null | References User.id |

**Table 2.6** Question

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique question identifier; auto-generated serial. |
| question_text | TEXT | — | Not Null | The full text content of the question as displayed to the student. |
| question_type | VARCHAR | 20 | Not Null | Format of the question: mcq (multiple choice), truefalse , or identification (short answer). |
| question_order | INTEGER | 4 | Not Null | Numeric ordering value that determines the display sequence of questions within an assessment. |
| assessment_id | BIGINT | 8 | Foreign Key Not Null | References Assessment.id |
| skill_category_id | BIGINT | 8 | Foreign Key Nullable | References SkillCategory.id |

**Table 2.7** AnswerChoice

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique answer choice identifier; auto-generated serial. |
| choice_text | TEXT | — | Not Null | The text content of the answer option presented to the student. |
| is_correct | BOOLEAN | 1 | Not Null | Flag indicating whether this choice is the correct answer |
| question_id | BIGINT | 8 | Foreign Key Not Null | References Question.id |

**Table 2.8** StudentResponse

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique student response session identifier; auto-generated serial. |
| started_at | TIMESTAMPTZ | — | Nullable | Timestamp recorded when the student begins the assessment |
| submitted_at | TIMESTAMPTZ | — | Nullable | Timestamp of when the student submitted the assessment |
| retake_allowed | BOOLEAN | 1 | Not Null | Flag set by the instructor to permit the student to retake the assessment |
| student_id | BIGINT | 8 | Foreign Key Not Null | References User.id |
| assessment_id | BIGINT | 8 | Foreign Key Not Null | References Assessment.id |

**Table 2.9** ResponseAnswer

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique response answer identifier; auto-generated serial. |
| text_answer | TEXT | — | Nullable | Free-text answer provided by the student for identification-type questions. |
| response_id | BIGINT | 8 | Foreign Key Not Null | References StudentResponse.id |
| question_id | BIGINT | 8 | Foreign Key Not Null | References Question.id |
| selected_choice_id | BIGINT | 8 | Foreign Key Nullable | References AnswerChoice.id |

**Table 2.10** SkillScore

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique skill score record identifier; auto-generated serial. |
| raw_score | INTEGER | 4 | Not Null | Number of correct answers the student achieved within the given skill category. |
| max_score | INTEGER | 4 | Not Null | Total number of questions in this skill category for the given assessment (maximum possible score). |
| percentage | FLOAT8 | 8 | Not Null | Computed proficiency percentage: (raw_score / max_score) × 100 |
| student_id | BIGINT | 8 | Foreign Key Not Null | References User.id |
| assessment_id | BIGINT | 8 | Foreign Key Not Null | References Assessment.id |
| skill_category_id | BIGINT | 8 | Foreign Key Not Null | References SkillCategory.id |

**Table 2.11** Company

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique company identifier; auto-generated serial. |
| name | VARCHAR | 200 | Not Null | Official name of the partner company or OJT host organization. |
| address | JSONB | — | Nullable | Structured geographic address in JSON format containing province, city/municipality, and barangay fields. |
| location_lat | FLOAT8 | 8 | Nullable | Geographic latitude coordinate of the company |
| location_lng | FLOAT8 | 8 | Nullable | Geographic longitude coordinate of the company. |
| created_at | TIMESTAMPTZ | — | Not Null | Timestamp of when the company record was added to the system. |
| added_by_id | BIGINT | 8 | Foreign Key Nullable | References User.id |

**Table 2.12** Position

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique position identifier; auto-generated serial. |
| title | VARCHAR | 150 | Not Null | Job title or designation of the OJT position (e.g., "Junior Web Developer", "IT Support Intern"). |
| slots_available | INTEGER | 4 | Not Null | Number of OJT slots the company is currently offering for this position. |
| company_id | BIGINT | 8 | Foreign Key Not Null | References Company.id |

**Table 2.13** PositionRequirement

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique position requirement record identifier; auto-generated serial. |
| required_percentage | FLOAT8 | 8 | Not Null | Minimum proficiency percentage (0.00–100.00) required in the specified skill category for a student to qualify for this position. Forms the company's requirement vector used in cosine similarity computation. |
| position_id | BIGINT | 8 | Foreign Key Not Null | References Position.id |
| skill_category_id | BIGINT | 8 | Foreign Key Not Null | References SkillCategory.id |

**Table 2.14** Recommendation

| Field Name | Data Type | Field Length | Constraint | Description |
|---|---|---|---|---|
| id | BIGINT | 8 | Primary Key | Unique recommendation record identifier; auto-generated serial. |
| match_score | FLOAT8 | 8 | Not Null | AI-generated cosine similarity score (0.00–100.00) representing the compatibility between a student's skill profile vector and a company position's requirement vector. Higher values indicate a stronger match. |
| generated_at | TIMESTAMPTZ | — | Not Null | Timestamp of when the recommendation was computed and stored by the system. |
| student_id | BIGINT | 8 | Foreign Key Not Null | References User.id |
| position_id | BIGINT | 8 | Foreign Key Not Null | References Position.id |

**Technologies, Concepts, Theories**

This subsection highlights the main models, algorithms, and technologies used in developing SkillBridge system. It presents the process of processing student skills data from their collection and preprocessing to feature extraction and recommendation generation and describes the tools and frameworks that make up the technological background of the system.

***Data Collection***

Data about students' skills are collected using the digital testing platform incorporated in SkillBridge. Instructors create skill-based tests using the web interface by manually adding the test items or adding them in batch using Excel spreadsheet file. Every test item corresponds to some specific competency category, for example, Programming, Database, Networking, Web Development, or Cybersecurity, and uses one of three question types: Multiple Choice, True/False, and Identification. After submitting the test, the chosen answers and results of the encoding are saved in the database. This answer is used as input data for scoring and competency profile creation [15].

***Data Pre-Processing***

Before analysis, submissions will go through pre-processing in order to increase the precision of grading. Deterministic grading of Multiple Choice and True/False type questions is done based on the correct-answer property contained in the answer choice records. Text normalization is applied in the case of identification type questions. It involves eliminating any extra whitespace and

conversion of both the student's answer and correct answer to lower case before comparison. This means that answers like "CPU", "cpu", and " Cpu " will be considered the same. The purpose of pre-processing is to reduce errors caused by differences in formatting in order to increase the precision of the results. Pre-processing is an essential part of machine learning and classification processes because the quality of input data determines the output. [16] .

***Feature Extraction***

After the scoring, the system would generate skill score percentages for each category. Raw score is the number of correct responses in a certain skill category, max score is the number of questions in a skill category, and the percentage would be calculated as (raw_score / max_score)*100. These percentages would be normalized to fall between 0 and 1 and organized in a skill vector with one dimension for each skill category. As an illustration, consider the case of a student whose skill vector is [0.82, 0.55, 0.30, 0.70], which implies that he has scored 82% in Web Development, 55% in Database Management, 30% in Networking and 70% in Programming. The student's competency profile would be represented by this numeric vector [17].

***Natural Language Processing Model Selection***

Natural Language Processing (NLP) will be used in the SkillBridge approach to process the generated profiles of students' competencies and job positions descriptions in companies. Three pre-trained models were considered

in the study: spaCy en_core_web_sm, spaCy en_core_web_md, and Stanford Stanza English. These pretrained configurations process generated competency profiles and position descriptions before TF-IDF vectorization and cosine similarity. They are preprocessing configurations, not three different recommendation algorithms. No additional model training was performed.

Tokenization separates text into tokens; lemmatization reduces inflected forms; and stopword filtering removes selected common terms. Qi et al. [32, Sec. 4, Tables 2 and 4] provide a research basis for considering preprocessing quality and speed together. Their linguistic tasks and toolkit versions differ from SkillBridge, so their results cannot directly select the best configuration for this system. Official pipeline capabilities are documented separately in [18].

Selection must follow the identified SkillBridge comparison run. The reported downstream accuracy is equal across configurations, while manuscript and saved-notebook timing values differ. The fastest-model conclusion therefore remains provisional until the run conditions and repeated timing measurements are reconciled.

**TF-IDF for Skill Category Suggestion and Text Vectorization**

Term Frequency-Inverse Document Frequency represents term importance relative to a document collection. Salton and Buckley discuss term weighting for text retrieval [17]. SkillBridge uses TF-IDF for skill-category suggestion and text matching. Kara et al. [31, Abstract and Sec. 4] also use a TF-IDF/cosine baseline in job recommendation; this supports its relevance without establishing that it is universally best.

On the question of skill category recommendation, TF-IDF is used in the process of comparing the question formulation provided by an instructor with the list of skill-category terms. Thus, when the instructor asks the question, “What is the purpose of a primary key in SQL?”, the important terms are “primary key” and “SQL” and the suggested skill category is Database. It allows reducing the need to manually tag questions and makes the categorization of questions consistent throughout all assessments.

For the generation of recommendations, TF-IDF transforms the generated student competency profiles and company-position descriptions into numerical text vectors. The student competency profile is based on assessment results, competency level and skill tags. The company-position description is formed based on position title, skill percentage, skill-category tags, and position tags. Scikit-learn’s TfidfVectorizer is utilized to convert these textual representations into TF-IDF feature matrix [19].

The formula for TF-IDF is as follows:

$$\textbf{TF-IDF}(t, d) = \textbf{TF}(t, d) \times \textbf{IDF}(t),$$

where TF(t, d) stands for the frequency of term t in document d and IDF(t) stands for the inverse document frequency of term t in the set of documents.

***Matching Algorithm: Cosine Similarity***

SkillBridge uses cosine similarity for measuring the degree of agreement between the competency profile of students and the needs of the job openings in companies. Cosine similarity computes the angle between two vectors in an n-dimensional space; when two vectors are similar to each other, they have a higher similarity value, suggesting that the student's profile and the job need are well-aligned, while otherwise, they have lower similarity values [20].

Mathematically, it is expressed as:

$$\textbf{Cosine Similarity}(A, B) = (A \cdot B) / (||A|| \times ||B||),$$

where A – student vector, B – company-position vector, A · B – dot product of A and B, ||A|| and ||B|| – magnitude of A and B respectively.

The application of cosine similarity in SkillBridge is two-fold. First, it is used to compare the vector of student category-score with the vector of requirements for the position in question. For example, a student with high scores in Programming and Databases will be compared with positions requiring similar competencies in those areas. Second, cosine similarity measures the TF-IDF vector of the generated student competency profile and the TF-IDF vector of each generated company–position description. Thus, both numerical values and textual similarity based on NLP techniques are considered.

Moreover, location-based similarity is used as an auxiliary decision-making factor. Coordinates for students are taken from their profile, and

the coordinates for companies are taken from the company profile. The distance between the student and each company is measured and transformed into a location-based similarity score, where shorter distances get higher scores and vice versa. This is consistent with OJT practice, where proximity of students and companies is a relevant factor in the decision.

The final recommendation score is calculated using the following weighted formula:

**Final Recommendation Score = 60% Category-Score Similarity + 25% Rich NLP Similarity + 15% Location-Based Similarity**

Category-score similarity takes up the biggest weight since the primary criterion for OJT placement is competence of the student. Rich NLP similarity confirms the recommendation by comparing generated competency profiles with company-position descriptions. Location-based similarity is the pragmatic aspect of OJT placement.

***Recommendation Process Flow***

.The recommendation process will begin when a student completes the competency assessment. The system will automatically check the submitted responses and compute the student's percentage score for each skill category. These scores will be used to generate a structured skill vector and a rich competency profile. At the same time, the company-position requirements encoded by the administrator will be used to generate requirement vectors and rich position descriptions. The selected NLP model will preprocess the generated texts through tokenization, lemmatization, and stopword removal. TF-IDF will then convert the processed texts into numerical vectors.

After feature extraction, cosine similarity will be applied to compute NLP-based text similarity between student profiles and position descriptions. Cosine similarity will also be applied to compute category-score similarity between student skill vectors and position requirement vectors. Location-based similarity will be computed separately using the distance between student and company coordinates. Finally, the system will combine the three similarity scores using the weighted formula and rank company positions from highest to lowest final recommendation score. The top-ranked results will be displayed to students and authorized users as OJT placement recommendations.

### *Technologies Used in the System*

#### React 19 with Vite - Frontend Framework

SkillBridge has been built using React 19, a modern JavaScript framework for building component-based web interfaces using Vite in order to make development process fast and produce efficient builds for production. Tailwind CSS v4 has been used to style the frontend and the application has been deployed using Vercel. React allows building sixteen different pages that are going to have role-based access, dashboards, and dynamic updates of the user interface without full page reloading. The choice of React is made based on its component reusability, large developer community, and native state management support required by a multirole web application [21].

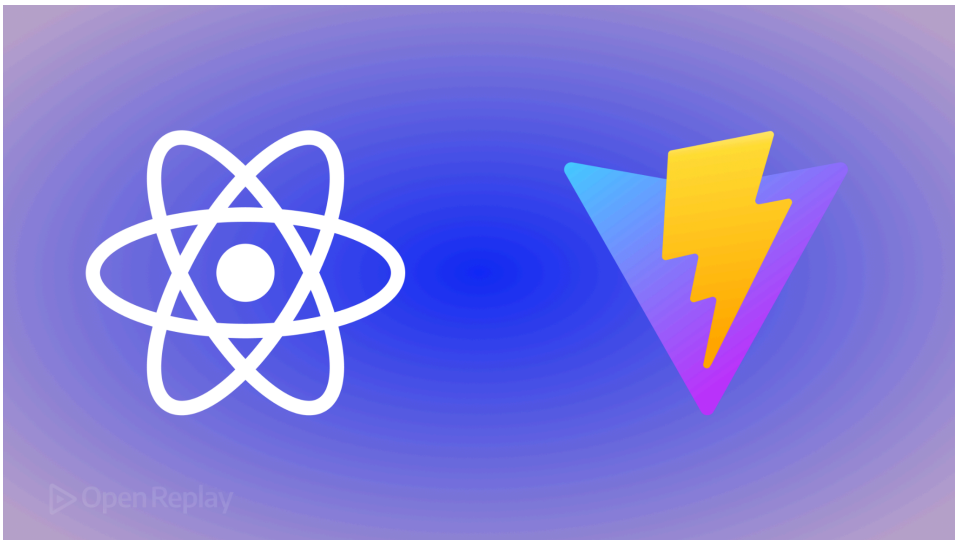



**Figure 11.** React - Frontend Framework.

**Python with Django and Django REST Framework - Backend Framework**

SkillBridge's server side code is going to be developed with the help of Python programming language together with Django framework 6.0.4 using the Model-View-Template architecture. Django REST framework 3.17.1 will provide the API interface and thus the RESTful endpoints which can be consumed from React front-end side. The decision to choose Python for backend development has been taken on account of its support for scikit-learn, the NLP algorithms library [22].




**Figure 12.** Django and Django REST Framework - Backend.

**PostgreSQL via Supabase - Database Management System**

The database management system that will be used is PostgreSQL where all the data will be stored in 14 tables of SkillBridge. The database will be run on Supabase where there is a provision of a managed PostgreSQL server with a graphical user interface for managing tables, daily backups, and IPv4 compatible connection pooling. The reason for choosing PostgreSQL is due to its good support of JSON fields in addresses, multi-table queries, and foreign key constraints [23].



**Figure 13.** PostgreSQL via Supabase - Database.

**spaCy, scikit-learn and NumPy - NLP and Machine Learning Libraries**

In the experiment, the performance of three pretrained NLP models was tested – spaCy `en_core_web_sm`, spaCy `en_core_web_md`, and Stanford Stanza English. Those models were utilized during the model comparison stage to tokenize, lemmatize, and remove stopwords from generated student competency profiles and company-position descriptions. spaCy is a Python NLP library with trained language pipelines capable of performing NLP preprocessing operations including tokenization and linguistic annotation [18]. Following the results of comparative testing, spaCy `en_core_web_sm` was chosen as the key NLP model to use in SkillBridge, as it provided the same level of recommendation accuracy as the other two NLP models but with less processing time.

For this research work, the scikit-learn library is used as the main library of the Python language for the implementation of TF-IDF feature extraction, cosine similarity calculation, and performance evaluation measures. In particular, the TfidfVectorizer is used for transformation of the generated competency profiles and company-position description into numerical vectors, and the cosine_similarity is used to calculate the similarity between the student profile and company-position description. The NumPy library is used for constructing and manipulating numerical vectors, including student skill vector, company-position description vector, and location similarity vector.




**Figure 14.** spaCy, scikit-learn and NumPy

**JWT Authentication - djangorestframework-simplejwt**

In order to achieve stateless authentication in SkillBridge, JSON Web Tokens (JWT) will be used. After authenticating, the backend server will provide an access token with a lifetime of eight hours and a refresh token with a lifetime of seven days. The access token and refresh token will be stored in localStorage in the frontend and will be sent with every API request using Bearer token authorization. The above strategy is both secure and stateless and works perfectly with the cross-origin configuration of the system; namely, a React frontend hosted on Vercel and a Django backend on Railway [24].




**Figure 15.** JWT Authentication

**Google OAuth 2.0 - Student Authentication**

Student authentication will be done through the Google account of the institution through Google OAuth 2.0. On the frontend side, the @react-oauth/google package is used to get the Google access token, which is sent to the backend side for validation on googleapis.com/oauth2/v3/userinfo. There is also a restriction on the domain, allowing only the @dnsc.edu.ph domain emails to create an account and login to the system. [25]




**Figure 16.** Google OAuth 2.0

**Leaflet with Nominatim - Map and Geocoding**

The interactive map on the student dashboard and the admin companies page will be created with Leaflet via CDN. For geocoding addresses to convert them to geographic coordinates (GPS), which will be kept in the database, the free Nominatim geocoding API that does not need any API key will be used. By calculating the great-circle distance from a student's address to companies with the help of Haversine formula, the frontend helps to estimate travel possibilities for students while assessing their placement [26].




**Figure 17.** Leaflet

**PSGC API - Philippine Address Data**

The API for the Philippine Standard Geographic Code (PSGC) would provide the official cascading drop-down lists of addresses based on the province, city or municipality, and barangay. The said drop-down lists would be used in creating student profiles and registering companies so that all address information in the system would be in the proper geographic format. Address information from the PSGC API would be saved into sessionStorage to avoid calling the API more than once [27].




**Figure 18.** PSGC API

**SheetJS (XLSX) - Excel File Parsing**

The SheetJS library will be included using a Content Delivery Network (CDN) and used to parse Excel files on the client side in the process of generating many assessments. The instructor will load an Excel file containing a set of questions that follow a specific structure and contain information such as question text, question type, choices, correct answer, and associated skills category. There will be a preview of the questions before uploading them into the system, and errors at the row level will be detected [28].

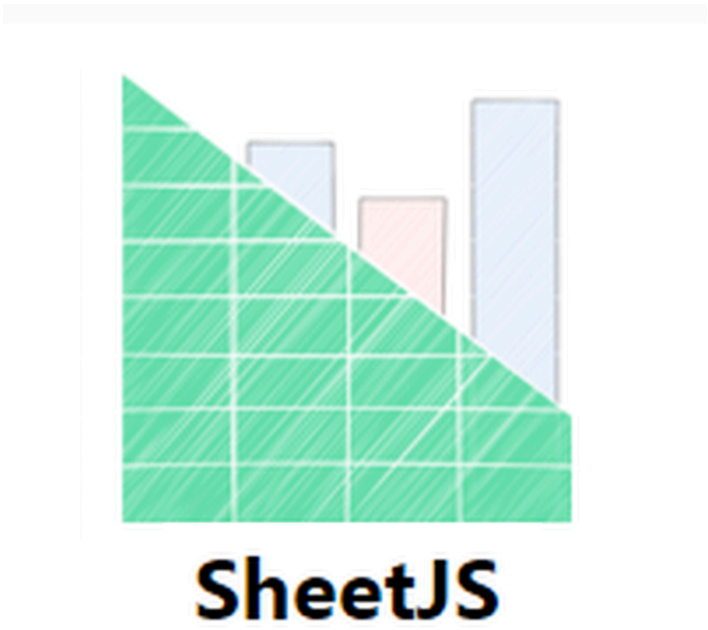



**Figure 19.** SheetJS

## SYSTEM TESTING AND IMPLEMENTATION

Testing examines whether SkillBridge integrates authentication, profile management, assessment creation and submission, automated scoring, competency profiling, text preprocessing, recommendation generation, location matching, dashboards, and reports. Application functionality and usability are evaluated separately from recommendation quality. Deployment preparation includes configuration, database setup, account provisioning, and user orientation.

### Results of Testing NLP Models and Evaluation Metrics

The notebook experiments compared spaCy en_core_web_sm, spaCy en_core_web_md, and Stanford Stanza English as pretrained preprocessing configurations. The resulting text was represented through TF-IDF and ranked by cosine similarity. This follows the approved comparison scope; it does not evaluate three separately trained placement classifiers.

Qi et al. [32, Sec. 4] compare linguistic annotation quality and runtime across NLP toolkits. This supports considering efficiency alongside quality, while differences in task, version, and hardware limit numerical transfer. Kara et al. [31, Abstract and Secs. 2.1 and 4] provide a related skill-based matching framework and a TF-IDF/cosine baseline. Neither study establishes SkillBridge accuracy or a preferred preprocessing configuration.

### Evaluation Design and Metric Definitions

The notebook evaluation uses 300 synthetic student profiles, six candidate positions, and one expected role assigned through each synthetic archetype. Its labels represent the simulation design, not independent judgments or observed placement outcomes. Role-related terms inferred from assessment scores also enter notebook student text, and the evaluated cohort participates in TF-IDF corpus construction. No independent final cohort reserved from design selection is documented.

Top-1 accuracy measures whether the first-ranked position matches the expected role. Hit Rate@3, called Top-3 accuracy in the original notebooks, measures whether that role appears among three suggestions. Precision, recall, and F1 are macro averages of first-ranked role predictions; they are not Precision@3, Recall@3, or proof that every suggested role is suitable. The confusion matrix identifies recurring errors.

Gunawardana and Shani [33, Secs. 4.2, 5.2 and 5.4] relate experimental design and evaluation measures to the recommendation task. Their shortlist discussion supports evaluating the displayed candidates; it does not prescribe SkillBridge macro averaging or a passing percentage. Complete suitability judgments would be needed to assess Precision@3 beyond the current single expected-role labels.

**Table 4 Comparative Analysis of NLP Configurations**

| NLP configuration | Top-1 | Hit Rate@3 | Macro precision | Macro recall | Macro F1 | Reported batch s |
|---|---|---|---|---|---|---|
| spaCy small | 93.00% | 99.67% | 0.9316 | 0.9302 | 0.9304 | 5.923 |
| spaCy medium | 93.00% | 99.67% | 0.9316 | 0.9302 | 0.9304 | 6.115 |
| Stanza English | 93.00% | 99.67% | 0.9316 | 0.9302 | 0.9304 | 154.567 |

Times are retained from the supplied manuscript and describe reported batch computation, not measured website latency. Macro metrics are fractions. The dataset has 300 synthetic cases and six candidates.

All configurations produced the same recorded Top-1 accuracy and Hit Rate@3 in this experiment. This is agreement with synthetic expected-role labels, rather than evidence of actual placement suitability or statistical equivalence across all possible data.

Qi et al. [32, Sec. 4, Speed comparison and Table 4] support examining runtime trade-offs. Their study does not choose between current spaCy small and medium for SkillBridge. The manuscript timings above differ from the attached rich-text notebook outputs; consequently, a final model-selection claim requires an identified run, stated hardware and versions, and repeated measurements.

### Outcome of Evaluating the Final Recommendation Model

The hybrid experiment adds category-score and location similarity to text matching. Category vectors summarize assessed percentages, text similarity compares generated profiles and descriptions, and location similarity represents geographic proximity. These components address different matching considerations.

### Weighted Hybrid Recommendation Rationale

Burke [29, Sec. 3.1] describes weighted hybrids that numerically combine component recommendation scores. This supports the score-combination architecture. It does not prescribe SkillBridge components or coefficients. Reference [30] remains a broader hybrid-recommender review.

**Final score = 0.60 C + 0.25 T + 0.15 L**

C is category-score similarity; T is NLP-assisted text similarity; L is location similarity. The component scores are expressed on a 0-to-1 scale before percentage display.

The implemented coefficients express the project design priority: assessed competency receives the largest coefficient, text alignment supplements that evidence, and proximity supplies a practical geographic consideration. This is the stated design rationale, not a weight set derived from Burke or an experimentally established optimum.

Stronger coefficient justification requires a sensitivity comparison on identical inputs, candidate positions, and labels, varying both text and location weights and including component-removal controls. Such validation is not claimed by this revision. Common numeric ranges do not ensure equal score distributions or equal effects on ranking.

The hybrid is evaluated using Top-1 accuracy, Hit Rate@3, macro precision, macro recall, macro F1, and computation time. Gunawardana and Shani [33] support task-appropriate comparison design; they do not validate these exact weights. Similarity scores are not calibrated probabilities of placement success.

**Table 5 Final Hybrid Recommendation Testing Findings**

| NLP configuration | Top-1 | Hit Rate@3 | Macro precision | Macro recall | Macro F1 | Reported batch s |
|---|---|---|---|---|---|---|
| spaCy small | 89.33% | 100.00% | 0.8970 | 0.8929 | 0.8936 | 6.092 |
| spaCy medium | 89.33% | 100.00% | 0.8970 | 0.8929 | 0.8936 | 6.114 |
| Stanza English | 89.33% | 100.00% | 0.8970 | 0.8929 | 0.8936 | 159.761 |

Original manuscript values are retained. Timing reconciliation is pending; these numbers must not be mixed with timings from another run. Macro metrics use first-ranked predictions.

The hybrid achieved 89.33% Top-1 accuracy and 100% Hit Rate@3 against synthetic expected roles, with macro F1 of 0.8936. The expected role was therefore in the shortlist for every case; this does not show that all three recommendations were suitable. The NLP-only experiment recorded 93% Top-1 accuracy, so these experiments do not establish that hybridization improved first-choice accuracy.

The saved final-hybrid notebook records different batch times and a different fastest spaCy configuration from this manuscript table. Neither table alone establishes a consistently fastest model. Repeated runs, timer boundaries, actual model identifiers, hardware, and software versions must be reconciled before final selection. This is a project evidence gap, not something an external citation can resolve.

Notebook computation excludes model loading and application/database/network operations; its scoring timer also includes feature construction. Partial batch computation must be distinguished from per-instance model inference and end-to-end application response time.

These preliminary findings support further evaluation of the matching approach. They do not establish superior real-world placement outcomes, optimal hybrid coefficients, or a universally best preprocessor. Runtime model configuration and the final research selection should be reconciled explicitly; a configured default is not selection evidence.

**System Test Plan**

In order to provide for the validation process of the SkillBridge software, a well-detailed System Test Plan is going to be prepared. This plan outlines the testing objectives, the scope of testing, the testing environment, the test cases, and the acceptance criteria ensuring that testing is carried out systematically based on the system requirements. The major objective of this phase is to validate that the integrated OJT placement decision-support platform works properly and effectively in real-world usage conditions.

The scope of testing involves validation of: user authentication for students, login for instructors and administrators, role-based access control, student profile management, assessment creation, bulk question upload, submission of student assessments, automatic scoring, skill categories computation, rich competency profiles creation, detailed company-position descriptions creation, NLP preprocessing, recommendations generation, location matching, dashboard rendering, and report generation.

For the testing environment, there will be used standard laptops and workstations with contemporary web browsers like Google Chrome and Microsoft Edge. For the backend, Railway service will be used, for the frontend – Vercel and for the database – Supabase PostgreSQL. The test data will include the student records, assessment questions, answer options, responses, skill categories, company records and coordinates, position records and requirements following the same structure as the actual database of SkillBridge.

Several test cases will be performed in order to test the normal behavior and edge cases. The test cases include: student login with DNSC Google account, disallowing non-DNSC email addresses, login of instructor with pending approval, assessment submission, finding answer normalization, category score calculation, rich profiles creation, recommendations ranking, distance calculations, company maps rendering, dashboard access, and blocking of unauthorized routes. The acceptance criteria require correct authentication of users, role checking, correct assessment scores, recommendations generation, correct company locations rendering and report generation without errors.

**Implementation Activities**

The implementation stage involves the deployment, configuration, and preparatory activities required for the user testing and deployment of SkillBridge. After developing and validating the components of the system internally, the backend application will be deployed through Railway, the frontend application through Vercel, and PostgreSQL database on Supabase. The Django migration

will be run to create the necessary database schema that includes tables for users, batches, skill categories, assessments, questions, student responses, skill scores, companies, positions, position requirements, and recommendations.

After deployment, the research team will configure the required environment variables that will include database credentials, JWT secret key, Google OAuth credentials, allowed hosts, and CORS settings. The initial account of an administrator will be created to manage user approval, company details, skill categories, and configuration.Testing will be made easier by creating sample data that includes company information, OJT positions, requirements for positions, assessments, and skill tags. Interoperability will be verified through user authentication, making API requests, submission of assessments, scoring, recommendation generation, mapping, and the use of the dashboard.

The activities of the implementation will also involve the creation of user accounts for the students, instructors, and administrators. The instructors will be instructed on how to create assessments, add questions, manage student batches, and review competency results. The students will be instructed on how to log in using their institutional Google accounts, complete their assessments, view their competency profiles, and see the ranking of OJT recommendations. The administrators will be instructed on how to manage company information, position requirements, OJT slots, and system analytics.

**User Acceptance Evaluation**

User Acceptance Evaluation is going to be performed to determine the level at which the needs and expectations of users of SkillBridge are met. This will include certain students, instructors, and administrators executing assigned tasks based on their positions. The student group will check account login, profile filling, assessment answering, competency profiles viewing, recommendations viewing, and the company map interaction. The instructors will test the functions of assessment creation, questions upload, management of student groups, scoring checking, and groups analytics. The administrators will test company management, position requirements encoding, recommendations monitoring, slots availability verification, and report creation.

At the end of performing the assigned tasks, participants will evaluate the system using a structured questionnaire based on selected ISO/IEC 25010 software quality characteristics, specifically functional suitability, usability, performance efficiency, reliability, and security. Descriptive statistics, including frequency, weighted means, and interpretations will describe the answers of the users. The findings of User Acceptance Evaluation will decide whether the system is acceptable for its users and help improve it before the final launch or any further development.

The acceptance criteria of the User Acceptance Evaluation will require users to perform their tasks without any critical errors, obtain correct scoring and recommendation results, and get an acceptable evaluation score regarding the

selected scale of interpretation. Feedback from users will be used for further interface improvement, workflow improvement, defect fixing, and enhancement of recommendations and reports of SkillBridge.

## SYSTEM MAINTENANCE

System Maintenance stage focuses on ensuring that SkillBridge is in an operational state and is reliable and improved after implementation. The phase involves monitoring system performance, troubleshooting of issues and applying necessary changes to improve and enhance the functioning of the application. Some of the tasks during this stage will include regular system checkups, updating software parts, and optimization of databases among others.

### System Security Plan

In order to ensure security of the system and its resources, the System Security Plan will be developed for SkillBridge based on ISO/IEC 27001 guidelines. The system will use authentication and role-based access control in order to prevent any unauthorized access to the system and its features. Data transfers from client to the server side will be secured using HTTPS protocol, and credentials and tokens storage will be protected using environment variables management and JWT signing.

**Table 6.** System Security Plan (ISO/IEC 27001).

| Security Domain (ISO 27001) | Control Area | Implementation in SkillBridge |
|---|---|---|
| A.5 Information Security Policies | Security Policy | The system will implement formal security policies to guide data protection, access control, and proper system usage throughout the application. |
| A.6 Organization of Information Security | Roles and Responsibilities | Three user roles — Admin, Instructor, and Student — will be clearly defined with distinct access privileges enforced through JWT claims and frontend route guards. |
| A.9 Access Control | User Access Management | Role-Based Access Control (RBAC) will be applied across all API endpoints. All frontend routes will be protected by PrivateRoute components that check user roles before granting access. |
| A.9 Access Control | Authentication Mechanism | JWT access tokens with an 8-hour validity and refresh tokens with a 7-day validity will be used. Google OAuth will handle student login, while email and password will be used for instructor and admin accounts. |
| A.10 Cryptography | Data Encryption | All data transmission will be secured over HTTPS, enforced by both Vercel and Railway. JWT tokens will be signed using a SECRET_KEY stored as a server-side environment variable. |
| A.12 Operations Security | Data Processing Integrity | Input validation will be applied to all API endpoints. Django form validation and unique database constraints will prevent invalid or duplicate data entry. |
| A.12 Operations Security | Logging and Monitoring | Railway and Vercel will provide deployment and request logs. Django |

| | | |
|---|---|---|
| | | request logging will also be configured to monitor system behavior and detect anomalies. |
| A.13 Communications Security | Network Security | CORS settings will be restricted to authorized origins only, preventing unauthorized cross-origin API access. |
| A.9 Access Control | Rate Limiting | The login endpoint will be throttled to 10 requests per minute. General user API endpoints will be limited to 200 requests per minute using DRF throttling classes. |
| A.17 Business Continuity | Data Backup and Recovery | Supabase will provide automated daily PostgreSQL backups to ensure data availability and recovery capability in the event of a system failure. |
| A.16 Incident Management | Incident Response | Procedures will be established to detect, document, and resolve security incidents, including system isolation and recovery steps as needed. |
| A.14 System Acquisition | Secure Development | All system secrets will be stored as environment variables and will never be hardcoded. Input validation and secure JWT configuration will be applied throughout the codebase. |

**System Maintenance Plan**

There will be a development of a System Maintenance Plan for directing the support and improvement of SkillBridge. The plan will include maintenance tasks that will be consistent with the ISO/IEC 25010 quality attributes. These include reliability, efficiency, security, and adaptability. The plan will include the process involved in monitoring, tuning, security patches, bug fixing, and management of the database.

**Table 7.** System Maintenance Plan (ISO/IEC 25010).

| ISO 25010 Attribute | Maintenance Activity | Description | Frequency |
|---|---|---|---|
| Reliability | System Monitoring | Monitor Railway deployment status, API response rates, and database connection health on a regular basis. | Daily |
| Performance Efficiency | API Optimization | Review and optimize endpoints that exhibit slow response times and improve database query performance through indexing. | As needed |
| Usability | Interface Improvement | Update dashboard design and overall user experience based on feedback gathered during UAT and post-deployment evaluation. | Periodically |
| Security | Security Updates | Apply Django and DRF security patches, rotate the JWT SECRET_KEY, and review CORS configuration settings. | Regularly |
| Maintainability | Code Refactoring and Debugging | Fix identified bugs and refactor key modules such as views.py and scoring.py to improve readability and long-term maintainability. | As needed |
| Compatibility | System Updates | Ensure compatibility with updated browser versions, React releases, and Tailwind | Periodically |

| | | | |
|---|---|---|---|
| | | CSS updates as they become available. | |
| Reliability | Database Backup Verification | Verify that Supabase automated backups are running correctly and test database restoration procedures. | Monthly |
| Performance Efficiency | Database Optimization | Clean up orphaned records and add indexes on frequently queried foreign key columns to maintain query performance. | Weekly |
| Scalability | Capacity Planning | Monitor Railway resource utilization and plan infrastructure upgrades if the student population or data volume grows significantly. | Quarterly |

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Reference numbers [1]-[30] are preserved from the supplied manuscript. Burke remains [29]. The added sources support methodological choices within their stated scope; none prescribes the exact 60/25/15 coefficients or certifies a SkillBridge model-selection result.